Assignment No. 2

Linux Log Analysis & Monitoring using Bash Scripts and Cron Jobs

1. Objective :

To configure log rotation, analyze Linux system logs, monitor disk usage, and automate monitoring tasks using Bash scripting and cron jobs on an Ubuntu EC2 instance.

2. Tools & Technologies Used :

- AWS EC2
- Ubuntu Linux
- Bash Scripting
- Cron Jobs
- Logrotate
- SSH

3. Tasks Performed :

Task 1: Created Application Log File
Created a sample application.log file inside the assignment2 directory.

```
admin-user@ip-172-31-40-180:~$ mkdir assignment2


admin-user@ip-172-31-40-180:~/assignment2$ cat application.log
Application started
User login successful
Database connected
admin-user@ip-172-31-40-180:~/assignment2$
```

Task 2: Configure Logrotate

Created logrotate configuration file to:

- Rotate logs weekly
- Keep logs for 4 weeks
- Compress old logs

Commands Used

sudo nano /etc/logrotate.d/application-log
sudo logrotate -f /etc/logrotate.d/application-log

```
admin-user@ip-172-31-40-18(
  GNU nano 7.2                       /etc/logrotate.d/application-log
/home/admin-user/assignment2/application.log {
    weekly
    rotate 4
    compress
    missingok
    notifempty
    su admin-user admin-user
}

                               [ Wrote 8 lines ]
^G Help        ^O Write Out   ^W Where Is    ^K Cut         ^T Execute     ^C Location    M-U Undo
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify     ^/ Go To Line  M-E Redo
```

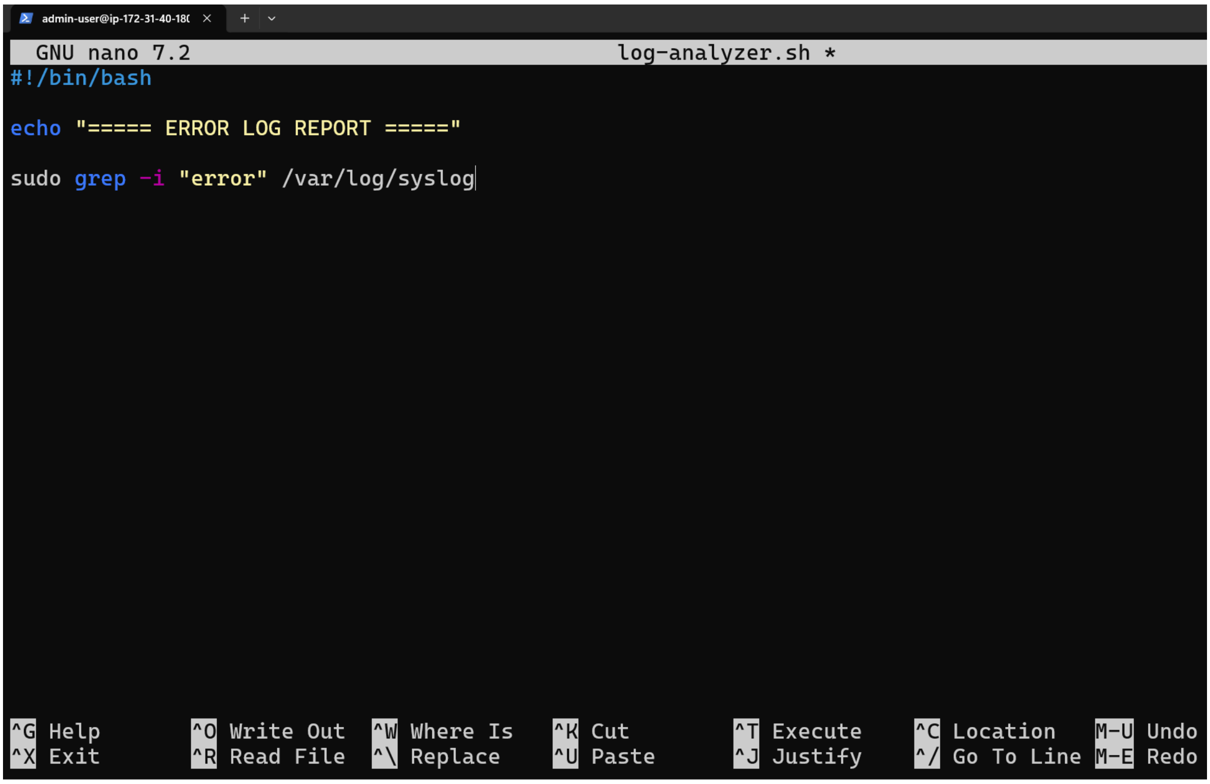
```
admin-user@ip-172-31-40-180:~/assignment2$ sudo nano /etc/logrotate.d/application-log
admin-user@ip-172-31-40-180:~/assignment2$ cat nano /etc/logrotate.d/application-log
cat: nano: No such file or directory
/home/admin-user/assignment2/application.log {
    weekly
    rotate 4
    compress
    missingok
    notifempty
    su admin-user admin-user
}
admin-user@ip-172-31-40-180:~/assignment2$ sudo logrotate -f /etc/logrotate.d/application-log
admin-user@ip-172-31-40-180:~/assignment2$ ls -l
total 4
-rw-rw-r-- 1 admin-user admin-user 78 May  6 16:02 application.log.1.gz
admin-user@ip-172-31-40-180:~/assignment2$
```

Task 3: Created Log Analyzer Script

Created log-analyzer.sh script to scan /var/log/syslog for errors using grep.

Purpose

To monitor and identify system errors from Linux logs.



chmod +x log-analyzer.sh

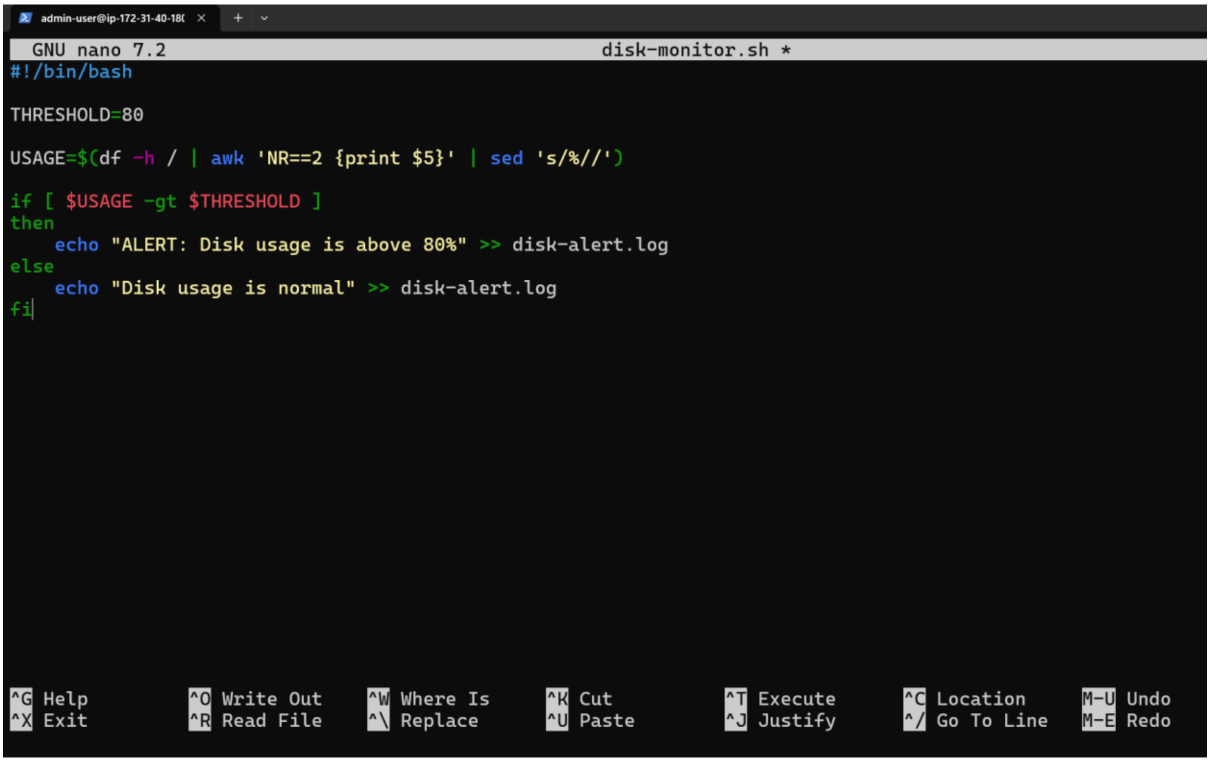
sudo ./log-analyzer.sh

```
admin-user@ip-172-31-40-180:~/assignment2$ nano log-analyzer.sh
admin-user@ip-172-31-40-180:~/assignment2$ ./log-analyzer.sh
===== ERROR LOG REPORT =====
2026-05-06T15:38:48.637112+00:00 ip-172-31-40-180 amazon-ssm-agent.amazon-ssm-agent[525]: Error occurred fetching the seelog config file path:  open /etc/amazon/ssm/seelog.xml: no such fil
e or directory
2026-05-06T15:38:48.653855+00:00 ip-172-31-40-180 amazon-ssm-agent.amazon-ssm-agent[525]: 2026-05-06 15:38:48.6394 WARN Error adding the directory '/etc/amazon/ssm' to watcher: no such fil
e or directory
2026-05-06T15:38:50.530088+00:00 ip-172-31-40-180 amazon-ssm-agent.amazon-ssm-agent[525]: 2026-05-06 15:38:48.8677 WARN EC2RoleProvider Failed to connect to Systems Manager with instance p
rofile role credentials. Err: retrieved credentials failed to report to ssm. Error: EC2RoleRequestError: no EC2 instance role found
2026-05-06T15:38:50.630457+00:00 ip-172-31-40-180 amazon-ssm-agent.amazon-ssm-agent[525]: 2026-05-06 15:38:48.9028 ERROR EC2RoleProvider Failed to connect to Systems Manager with SSM role
credentials. error calling RequestManagedInstanceRoleToken: AccessDeniedException: Systems Manager's instance management role is not configured for account: 339333478686
2026-05-06T15:38:50.731783+00:00 ip-172-31-40-180 amazon-ssm-agent.amazon-ssm-agent[525]: 2026-05-06 15:38:48.9029 ERROR [CredentialRefresher] Retrieve credentials produced error: no valid
 credentials could be retrieved for ec2 identity. Default Host Management Err: error calling RequestManagedInstanceRoleToken: AccessDeniedException: Systems Manager's instance management r
ole is not configured for account: 339333478686
2026-05-06T15:38:50.932239+00:00 ip-172-31-40-180 amazon-ssm-agent.amazon-ssm-agent[525]: 2026-05-06 15:38:48.9031 INFO [SerialPort] Write to serial port: SSM Agent unable to acquire crede
ntials: <error>no valid credentials could be retrieved for ec2 identity. Default Host Management Err: error calling RequestManagedInstanceRoleToken: AccessDeniedException: Systems Manager'
s instance management role is not configured for account: 339333478686
2026-05-06T15:38:50.932351+00:00 ip-172-31-40-180 amazon-ssm-agent.amazon-ssm-agent[525]: #011status code: 400, request id: 32661de0-529e-4212-bc94-e62e58480d6b</error>
2026-05-06T15:41:20.557374+00:00 ip-172-31-40-180 fwupd[1037]: 15:41:20.557 FuPluginLinuxSleep   could not open /sys/power/mem_sleep: Error opening file /sys/power/mem_sleep: No such file
or directory
2026-05-06T15:44:13.948958+00:00 ip-172-31-40-180 amazon-ssm-agent.amazon-ssm-agent[1552]: Error occurred fetching the seelog config file path:  open /etc/amazon/ssm/seelog.xml: no such fi
le or directory
2026-05-06T15:44:13.951639+00:00 ip-172-31-40-180 amazon-ssm-agent.amazon-ssm-agent[1552]: 2026-05-06 15:44:13.9491 WARN Error adding the directory '/etc/amazon/ssm' to watcher: no such fi
le or directory
2026-05-06T15:44:15.817878+00:00 ip-172-31-40-180 amazon-ssm-agent.amazon-ssm-agent[1552]: 2026-05-06 15:44:13.9803 WARN EC2RoleProvider Failed to connect to Systems Manager with instance
profile role credentials. Err: retrieved credentials failed to report to ssm. Error: EC2RoleRequestError: no EC2 instance role found
2026-05-06T15:44:15.918304+00:00 ip-172-31-40-180 amazon-ssm-agent.amazon-ssm-agent[1552]: 2026-05-06 15:44:14.0089 ERROR EC2RoleProvider Failed to connect to Systems Manager with SSM role
 credentials. error calling RequestManagedInstanceRoleToken: AccessDeniedException: Systems Manager's instance management role is not configured for account: 339333478686
2026-05-06T15:44:16.018636+00:00 ip-172-31-40-180 amazon-ssm-agent.amazon-ssm-agent[1552]: 2026-05-06 15:44:14.0090 ERROR [CredentialRefresher] Retrieve credentials produced error: no vali
d credentials could be retrieved for ec2 identity. Default Host Management Err: error calling RequestManagedInstanceRoleToken: AccessDeniedException: Systems Manager's instance management
role is not configured for account: 339333478686
2026-05-06T15:44:16.219363+00:00 ip-172-31-40-180 amazon-ssm-agent.amazon-ssm-agent[1552]: 2026-05-06 15:44:14.0093 INFO [SerialPort] Write to serial port: SSM Agent unable to acquire cred
entials: <error>no valid credentials could be retrieved for ec2 identity. Default Host Management Err: error calling RequestManagedInstanceRoleToken: AccessDeniedException: Systems Manager
's instance management role is not configured for account: 339333478686
2026-05-06T15:44:16.219453+00:00 ip-172-31-40-180 amazon-ssm-agent.amazon-ssm-agent[1552]: #011status code: 400, request id: c5149ad0-af7d-4cfa-8c71-bd18aba37d29</error>
2026-05-06T16:12:01.011620+00:00 ip-172-31-40-180 amazon-ssm-agent.amazon-ssm-agent[1552]: 2026-05-06 16:12:01.0110 WARN EC2RoleProvider Failed to connect to Systems Manager with instance
profile role credentials. Err: retrieved credentials failed to report to ssm. Error: EC2RoleRequestError: no EC2 instance role found
2026-05-06T16:12:01.111601+00:00 ip-172-31-40-180 amazon-ssm-agent.amazon-ssm-agent[1552]: 2026-05-06 16:12:01.0464 ERROR EC2RoleProvider Failed to connect to Systems Manager with SSM role
 credentials. error calling RequestManagedInstanceRoleToken: AccessDeniedException: Systems Manager's instance management role is not configured for account: 339333478686
2026-05-06T16:12:01.212001+00:00 ip-172-31-40-180 amazon-ssm-agent.amazon-ssm-agent[1552]: 2026-05-06 16:12:01.0464 ERROR [CredentialRefresher] Retrieve credentials produced error: no vali
d credentials could be retrieved for ec2 identity. Default Host Management Err: error calling RequestManagedInstanceRoleToken: AccessDeniedException: Systems Manager's instance management
role is not configured for account: 339333478686
2026-05-06T16:15:01.825066+00:00 ip-172-31-40-180 fwupd[1960]: 16:15:01.825 FuPluginLinuxSleep   could not open /sys/power/mem_sleep: Error opening file /sys/power/mem_sleep: No such file
or directory
```

Task 4: Created Disk Monitoring Script

Created disk-monitor.sh script to:

- Check disk usage
- Alert when usage exceeds 80%
- Save result in disk-alert.log



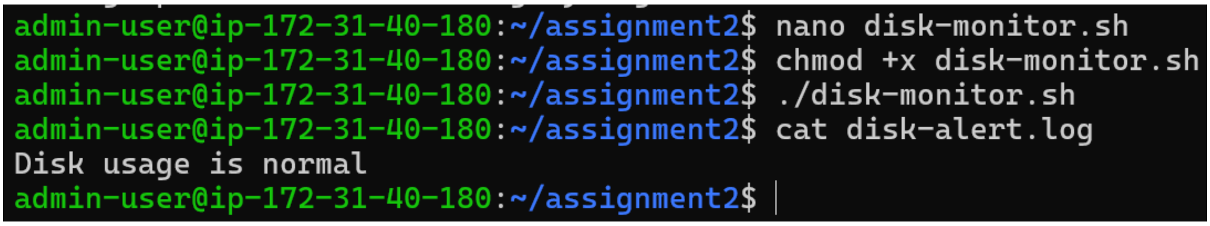
```
admin-user@ip-172-31-40-18(
GNU nano 7.2                                  disk-monitor.sh *
#!/bin/bash

THRESHOLD=80

USAGE=$(df -h / | awk 'NR==2 {print $5}' | sed 's/%//')

if [ $USAGE -gt $THRESHOLD ]
then
    echo "ALERT: Disk usage is above 80%" >> disk-alert.log
else
    echo "Disk usage is normal" >> disk-alert.log
fi

^G Help        ^O Write Out   ^W Where Is    ^K Cut         ^T Execute     ^C Location    M-U Undo
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify     ^/ Go To Line  M-E Redo
```



```
admin-user@ip-172-31-40-180:~/assignment2$ nano disk-monitor.sh
admin-user@ip-172-31-40-180:~/assignment2$ chmod +x disk-monitor.sh
admin-user@ip-172-31-40-180:~/assignment2$ ./disk-monitor.sh
admin-user@ip-172-31-40-180:~/assignment2$ cat disk-alert.log
Disk usage is normal
admin-user@ip-172-31-40-180:~/assignment2$
```

Disk usage was normal because the server usage was below 80%.

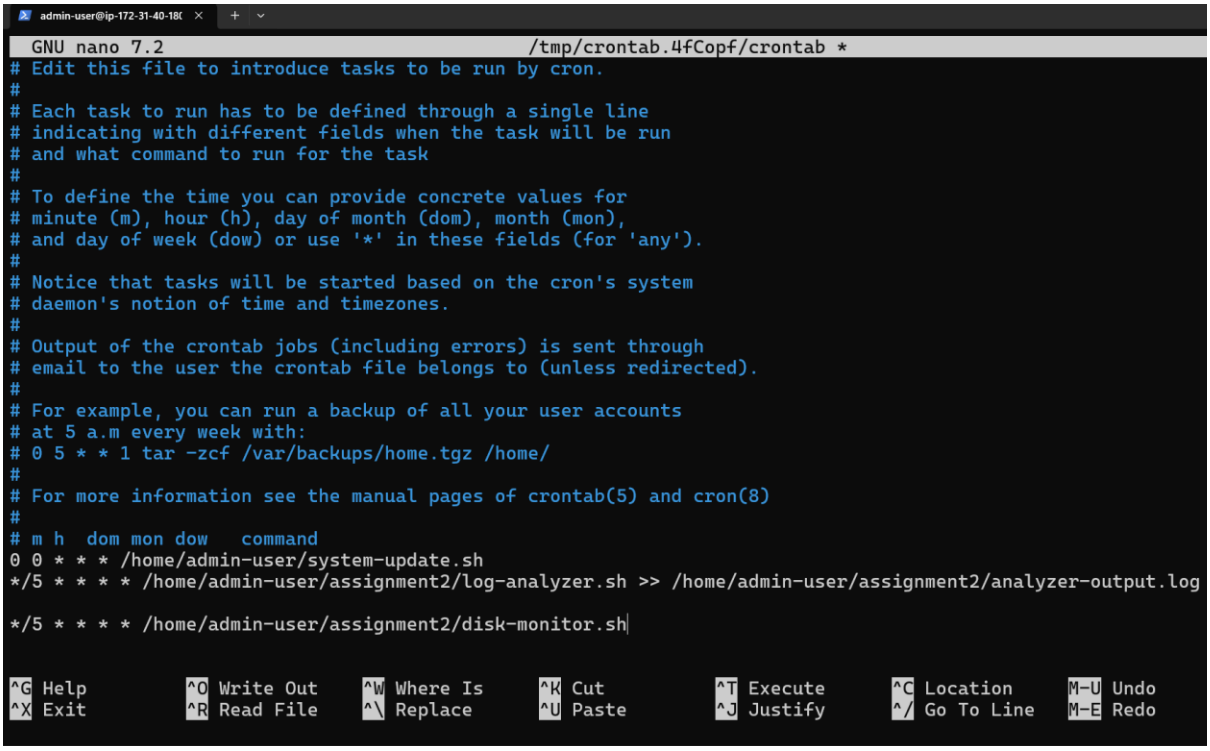
Task 5: Scheduled Scripts Using Cron

Configured cron jobs to run monitoring scripts every 5 minutes.

Cron Jobs Added

*/5 * * * * /home/admin-user/assignment2/log-analyzer.sh >> /home/admin-user/assignment2/analyzer-output.log

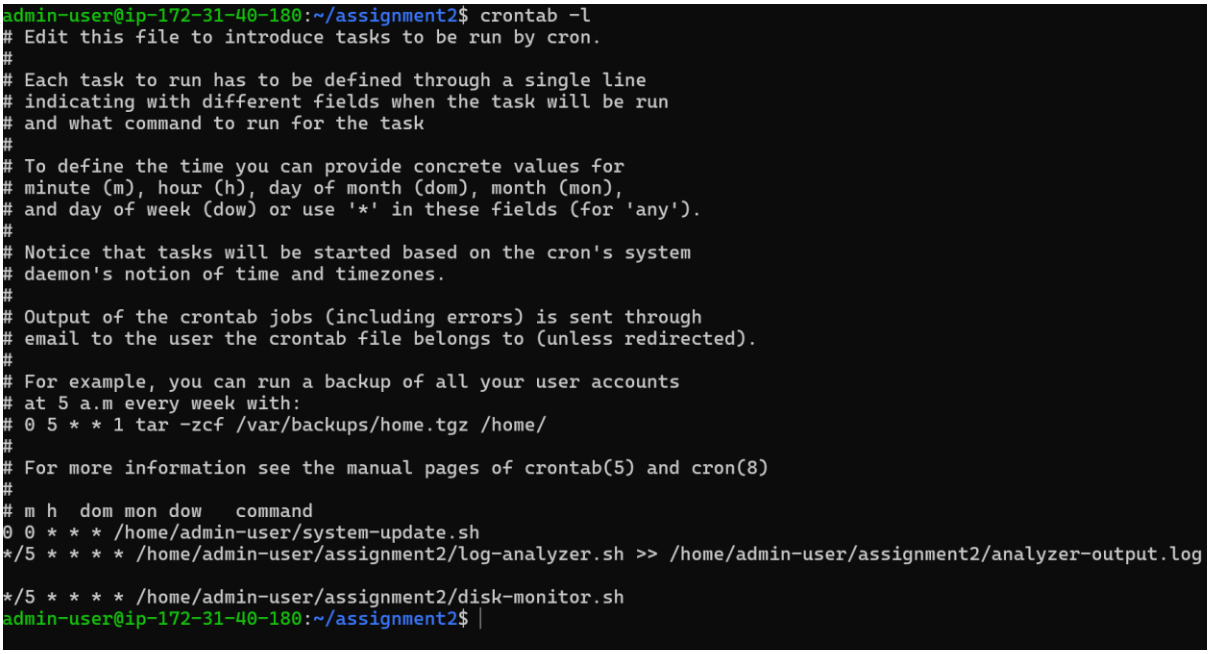
*/5 * * * * /home/admin-user/assignment2/disk-monitor.sh



```
admin-user@ip-172-31-40-18(
GNU nano 7.2                                    /tmp/crontab.4fCopf/crontab *
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h  dom mon dow   command
0 0 * * * /home/admin-user/system-update.sh
*/5 * * * * /home/admin-user/assignment2/log-analyzer.sh >> /home/admin-user/assignment2/analyzer-output.log

*/5 * * * * /home/admin-user/assignment2/disk-monitor.sh

^G Help        ^O Write Out   ^W Where Is    ^K Cut         ^T Execute     ^C Location    M-U Undo
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify     ^/ Go To Line  M-E Redo
```



```
admin-user@ip-172-31-40-180:~/assignment2$ crontab -l
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h  dom mon dow   command
0 0 * * * /home/admin-user/system-update.sh
*/5 * * * * /home/admin-user/assignment2/log-analyzer.sh >> /home/admin-user/assignment2/analyzer-output.log

*/5 * * * * /home/admin-user/assignment2/disk-monitor.sh
admin-user@ip-172-31-40-180:~/assignment2$
```

4. Output / Results :

- Log files rotated successfully
- Error monitoring script executed successfully
- Disk monitoring script generated alert logs
- Cron jobs scheduled automation successfully
- Monitoring scripts were automated successfully using cron scheduling.

5. Key Learnings :

- Learned Linux log management
- Understood logrotate configuration
- Learned Bash scripting basics
- Learned cron job automation
- Learned disk monitoring and alerting

6. Conclusion :

Successfully implemented Linux monitoring and automation using Bash scripts, logrotate, and cron jobs on an Ubuntu EC2 instance.